

Brussels, 19 May 2026

COST 059/26

DECISION

Subject: Memorandum of Understanding for the implementation of the COST Action “Simulation Education And Research Collaborative in Healthcare” (SEARCH) CA25145

The COST Member Countries will find attached the Memorandum of Understanding for the COST Action Simulation Education And Research Collaborative in Healthcare approved by the Committee of Senior Officials at its 227th meeting held in Brussels on 19 May 2026.

MEMORANDUM OF UNDERSTANDING

For the implementation of a COST Action designated as

COST Action CA25145

SIMULATION EDUCATION AND RESEARCH COLLABORATIVE IN HEALTHCARE (SEARCH)

The COST Members through the present Memorandum of Understanding (MoU) wish to undertake joint activities of mutual interest and declare their common intention to participate in the COST Action, referred to above and described in the Technical Annex of this MoU.

The Action will be carried out in accordance with the set of COST Implementation Rules approved by the Committee of Senior Officials (CSO), or any document amending or replacing them.

The main aim and objective of the Action is to build a sustainable, inclusive European network that advances healthcare simulation research and improvement science, reduces fragmentation, strengthens collaboration and knowledge exchange, and accelerates the translation of evidence into practice to improve care, health systems, and workforce development. This will be achieved through the specific objectives detailed in the Technical Annex.

The present MoU enters into force on the date of the approval of the COST Action by the CSO.

OVERVIEW

Summary

Healthcare simulation—techniques that recreate clinical events or environments for training, assessment, quality improvement, and research—has evolved from basic technical training into a well-established, complex, evidence-based approach that enhances patient safety and healthcare quality, as well as individual and team skill development.

However, simulation research in Europe remains fragmented across national, disciplinary, and professional boundaries, limiting collaboration, knowledge sharing, and equitable implementation. Five major challenges constrain its potential: keeping pace with rapid technological change; preparing healthcare professionals for increasingly complex and multicultural systems; translating simulation into measurable improvements in quality and safety; overcoming disciplinary and geographic fragmentation; and ensuring sustainable career pathways for emerging researchers.

The **Simulation Education And Research Collaborative in Healthcare (SEARCH)** will address these challenges by establishing a pan-European community of practice and research that connects academics, clinicians, industry, policymakers, scientific societies, and patient representatives. SEARCH will coordinate and strengthen healthcare simulation research, build capacity, and promote inclusiveness across Europe.

Building on the diffuse expertise of the European simulation community, SEARCH will align research efforts, strengthen capacity, and promote mutual learning, knowledge mobilisation, and translation. It will map existing resources, projects, and opportunities; identify barriers to knowledge transfer; develop an open-access toolkit alongside a ten-year European research agenda; foster interdisciplinary consortia; and help establish sustainable career pathways.

By aligning research, innovation, and policy, SEARCH will advance simulation science and establish a sustainable, inclusive European ecosystem for healthcare simulation research that will enhance healthcare outcomes across Europe well beyond the lifetime of the Action.

Areas of Expertise Relevant for the Action • Health Sciences: Health services, health care research	Keywords • Healthcare simulation • Interdisciplinary Research • Experiential learning • Healthcare Quality and Safety • Community of Practice
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Specific Objectives

To achieve the main objective described in this MoU, the following specific objectives shall be accomplished:

Research Coordination

- RCO 1. Build and foster a robust and sustainable pan-European network of health and care simulation research and innovation

This objective aims to overcome fragmentation by establishing a coordinated European community that connects researchers, practitioners, policymakers, and industry. It focuses on building lasting structures for

collaboration and leadership across sectors.

- RCO 2. Coordinate the generation, mobilisation, and application of knowledge

This objective seeks to strengthen the flow of knowledge across academia, industry, education, and healthcare, transforming expertise into accessible, actionable tools for the wider community.

- RCO3. Develop a research agenda for healthcare simulation for the next decade

This objective will scope, define, and develop a shared strategic direction for simulation research across Europe, aligning scientific, clinical, and industrial priorities with future healthcare needs.

Capacity Building

- CBO 1. Map infrastructure, expertise and resources to promote equitable access to resources, training, and knowledge

This objective aims to enhance transparency and accessibility by cataloguing Europe's simulation infrastructure, thereby promoting equity in research opportunities and collaboration.

- CBO2. Identify, document, and reduce systematic barriers to knowledge mobilisation and translation in healthcare simulation

This objective aims to identify and address cultural, linguistic, regulatory, and resource-related barriers encountered during knowledge transfer between academia, industry, and healthcare, that limit effective knowledge transfer across Europe.

- CBO3. Promote knowledge mobilisation and translation in healthcare simulation

This objective seeks to embed continuous learning and exchange through joint academic–industry–healthcare training and networking opportunities.

- CBO 4. Diversify the healthcare simulation research landscape through cross-disciplinary collaborations involving industry and disciplines outside healthcare

This objective will expand the field's reach by fostering partnerships with disciplines from non-health disciplines (e.g., engineering) and from industry.

- CBO 5. Establish a sustainable career trajectory for YRIs to develop the unique skills required to conduct innovative research in healthcare simulation

This objective aims to cultivate the next generation of simulation researchers through targeted training, mentorship, and mobility opportunities across Europe.

TECHNICAL ANNEX

1. EXCELLENCE IN S&T AND NETWORKING

A. Main challenge

Healthcare is a dynamic field with constant change at every level, from health systems to individual healthcare providers, who are required to understand and address epidemiological transitions and complex challenges. These include, for example: (a) Ageing populations: Many European countries, like most of the global north, face aging demographics with multiple co-morbidities, requiring more specialised care and healthcare resources. (b) Inequity of access: Health and social care workers and resources are unevenly distributed within and across geographic and political boundaries, limiting the support of this complex care. (c) Workforce shortages: There are shortages of appropriately trained healthcare workers, due to many factors. This impacts healthcare workers' health and wellbeing, with many colleagues leaving the workforce or experiencing burnout. (d) Innovation demands: New processes, systems, technologies and treatments are continually needed to redesign facilities, investigate, or mitigate acute and chronic health conditions and to treat existing and emergent pathogens with pandemic potential. Translational simulation and research can help healthcare address these complex challenges, improving and supporting healthcare systems, organisations and individuals.

However, the main challenge at present is the fragmentation of simulation research efforts across Europe, despite the widespread interest in exploring and identifying simulation as part of the solution to emerging healthcare problems. To manage the aforementioned dynamic healthcare challenges, the development of research structures which address the unique needs of Europe should be accelerated through advancing healthcare simulation research and improvement science in Europe. This highlights the need for an accessible, sustainable, and inclusive healthcare simulation research network that connects academics, healthcare practice, industry, and patient representatives across Europe. This overarching challenge of fragmentation manifests in several interconnected ways, each representing a critical area where progress is needed:

Challenge 1 (C1): Rapid Technological Advancement. New simulation technologies are emerging rapidly, with great potential, yet the technological landscape is moving too rapidly for healthcare educators and practitioners to understand, plan, invest in, and adopt these advancements. Technologies for virtual and immersive environments offer opportunities to revolutionise training for delivering complex patient care in a wide variety of settings. Currently, coordination between simulation research, industry, and healthcare practice is lacking. This lack can, in part, be explained through the extreme interdisciplinary nature of simulation which involves numbers academic, technological and industrial silos. This prevents adoption of rapid technological advancement and is compounded by existing inequalities across European healthcare contexts since many emerging simulation technologies are expensive to acquire and implement. *We need collaborative research teams to explore research questions, such as: Which simulation methodologies are required to tackle the dynamic and complex healthcare challenges and epidemiological transitions? How should new technologies like AI and immersive environments be integrated into healthcare training and practice? What standards will emerge, and how will those standards be shared and upheld across a diverse simulation ecosystem and across geographical boundaries? What new ethical challenges will emerge?*

Challenge 2 (C2): Educating Clinicians for Dynamic Healthcare. While simulation as an educational approach is now ubiquitous in affluent healthcare systems, we need more well-coordinated research to explain how best to use simulation to educate clinicians for complex dynamic healthcare and the specific needs of European contexts.

There is complexity in adopting simulation pedagogy due to: (a) issues of national, institutional, and professional healthcare culture; (b) language related to both professional jargon as well as multilingualism, especially in the mobile and diverse healthcare professionals (HCP) population in Europe (Ingels et al., 2022); (c) personal and cultural perspectives related to religion, geographical and political background, gender, and sexuality; d) equity of access to both formal and informal training opportunities, and (e) equity of access to digital technologies. This raises questions that urgently needed, larger scale, multi-centre programmes of research can help answer, but opportunities for this are rare and must be coordinated and supported. *We need collaborative research teams to explore research questions such as: what simulation technologies and approaches can help prepare HCPs from*

different linguistic backgrounds to work together? How can simulation help teams of clinicians from different countries to come together in an integrated way, providing the best care for patients?

Challenge 3 (C3): Enhancing Healthcare Quality and Safety Through Simulation. Simulation is a tool for diagnosing healthcare systems issues, for improving healthcare quality, and for creating safer and more patient-centred care, yet there is a paucity of this simulation research output from Europe.

Simulation for healthcare quality improvement is an emerging field in North America and Australia but has been slower to emerge in European contexts. While many causes can be identified, including the intensive service pressures in national health systems of European countries, using simulation to improve health system quality requires greater depth and capacity in the simulation research community to drive innovative research methods and approaches. *We need coordinated methodological support from diverse disciplines, such as implementation science, organisational behaviour, management, psychology, sociology, anthropology, coming together to work within healthcare simulation, for us to embrace and embed new practices.*

Challenge 4 (C4): Breaking Down Silos and Encouraging Collaboration. Healthcare simulation research and practice has developed in geographical, professional, and disciplinary silos, which leads to research findings which may not be translated or implemented in European contexts.

Europe has no infrastructure to promote collaboration for healthcare simulation research across professional disciplines and across national borders, or to coordinate the vision and agenda of research in healthcare simulation. This lack of infrastructure impacts the scope and scale of research possibilities; the coherency of transnational research agendas; neglects contextual and cultural factors; produces findings of limited generalisability. In North America, philanthropically funded research networks have helped both the development of individual researchers and the generation of specific research evidence such as INSPIRE (2022), which focuses on paediatric simulation. *We need research leadership, collaboration and coordination across Europe to generate new knowledge about the unique needs of healthcare in Europe, and to identify funding possibilities for healthcare simulation research, to create a harmonised, comprehensive, long-term, sustainable and inclusive simulation research agenda that aligns with patient safety and healthcare system priorities.*

Challenge 5 (C5): Nurturing a New Generation of Simulation Researchers. As a relatively nascent field, there is no well-developed pipeline for early career expertise development, no clear career trajectory for healthcare simulation researchers, and few opportunities for experts to move between research, industry, policymaking, and practice.

For example: Healthcare simulation researchers may come to the field with some or little prior research experience and from diverse fields including education, public health, clinical trials, or basic science. Healthcare simulation requires a range of research methods, including those from domains of science and engineering as well as social science. Some novice simulation researchers have extensive clinical experience but limited, if any, research background and exposure to healthcare simulation and/or scientific research, resulting in suboptimal research skills. *We need coordinated research agendas, drawing together colleagues from across the continent to pursue large, grant-funded, multi-site programmes of simulation research. Thoughtful coordination of training opportunities, such as a European Simulation Research Fellowship, will help to move the field onto a sustainable footing and provide opportunities for early-career simulation researchers, focused on meeting major healthcare challenges in Europe.*

B. Objectives

This Action objectives directly address the challenges outlined above and underpin our mission to foster collaboration, advance healthcare simulation research, and deliver tangible impact across European healthcare systems and organisations. Each objective is articulated using the SMART framework, with further detail in section 3B.

RESEARCH COORDINATION OBJECTIVES (RCO)

RCO 1. Build and foster a robust and sustainable pan-European network of health and care simulation research and innovation (Addresses: C1, C4, C5)

This objective aims to overcome fragmentation by establishing a coordinated European community that connects researchers, practitioners, policymakers, and industry. It focuses on building lasting structures for collaboration and leadership across sectors.

To do this, the Action will develop a blended model of face-to-face and online knowledge exchange, supported by a web-based platform and a self-governing Representative Council including Young Researchers and Innovators (YRIs) and representatives from COST Inclusiveness Target Countries (ITCs), by the end of Year 1 and reviewed annually. The Action aims to establish a long-lasting and diverse simulation research network, reaching at least 300 members by year 4, and ensure $\geq 60\%$ active participation in annual events. This network will therefore lay the groundwork for a sustainable, self-governing body that will guide and support the field beyond the lifetime of the Action.

RCO 2. Coordinate the generation, mobilisation, and application of knowledge (Addresses: C1, C2, C3)

This objective seeks to strengthen the flow of knowledge across academia, industry, education, and healthcare, transforming expertise into accessible, actionable tools for the wider community.

To do this, the Action will unite academia, industry, education, and clinical practice to identify gaps and best practices, capture multidirectional knowledge flows, and document them in a public toolkit. This toolkit will support sustained knowledge exchange beyond the lifetime of the Action, as it will be peer-generated, peer-led, and peer-sustained. The toolkit will be launched by Year 2, updated twice per year, with a success metric of ≥ 100 downloads by Year 4.

RCO3. Develop a research agenda for healthcare simulation for the next decade (Addresses: C1, C2, C3)

This objective will scope, define, and develop a shared strategic direction for simulation research across Europe, aligning scientific, clinical, and industrial priorities with future healthcare needs. The research agenda will give industry, academia, and healthcare a clear focus for knowledge generation through the next decade and beyond. Embedded in the research agenda will be a governance and sustainability plan, ensuring that the research agenda remains viable beyond the life of the Action and is revised and updated as technology and the healthcare landscape continue to develop.

To do this, the Action will co-create a long-term European research agenda with input from key stakeholders, including WHO, European societies, and public representatives, to be published by year 3. In addition, the Action will produce at least 2 white papers derived from the research agenda by year 4 and obtain endorsement from at least 3 European healthcare organisations.

CAPACITY-BUILDING OBJECTIVES (CBO)

CBO 1. Map infrastructure, expertise and resources to promote equitable access to resources, training, and knowledge (Addresses: C1, C2, C3)

This objective aims to enhance transparency and accessibility by cataloguing Europe's simulation infrastructure, thereby promoting equity in research opportunities and collaboration. The resource map will be developed in partnership with national and European societies, integrating their expertise and legacy to ensure continuity beyond the lifetime of the Action. Custodianship will be transferred to the Representative Council, which will oversee its ongoing maintenance in collaboration with partner societies. The resource map will form an integral component of the Action's web-based platform.

To do this, the Action will create a dynamic, extensible, sustainable, and open-access peer-coordinated resource map, documenting existent and emerging simulation centres, expertise, funding opportunities, technologies and policies, across Europe and beyond. The map will be launched by Year 1 with ≥ 50 simulation institutions and updated annually.

CBO2. Identify, document, and reduce systematic barriers to knowledge mobilisation and translation in healthcare simulation (Addresses: C2, C3)

This objective aims to identify and address cultural, linguistic, regulatory, and resource-related barriers encountered during knowledge transfer between academia, industry, and healthcare, that limit effective knowledge transfer across Europe. Practical strategies will be identified, providing guidance for future research collaborations.

To do this, the Action will develop and publish a comprehensive report outlining barriers and enablers to knowledge mobilisation and translation, along with mitigation strategies and case studies of good practice. The compilation will be published as part of the toolkit by the end of Year 3.

CBO3. Promote knowledge mobilisation and translation in healthcare simulation (Addresses: C1, C2, C3)

This objective seeks to embed continuous learning and exchange through joint academic–industry–healthcare training and networking opportunities.

To do this, the Action will deliver continuous professional development through regular networking and specialised training opportunities, including a minimum of 3 networking events, 4 workshops, 8 seminars, 4 training schools, 12 masterclasses, and 6 STSMs over the four-year period, supported by the network infrastructure developed under other objectives (such as CBO1).

CBO 4. Diversify the healthcare simulation research landscape through cross-disciplinary collaborations involving industry and disciplines outside healthcare (Addresses: C1, C4)

This objective will expand the field's reach by fostering partnerships with disciplines from non-health disciplines (e.g., engineering) and from industry. These partnerships will foster innovative, cross-sector collaborations that transcend traditional disciplinary boundaries, offering novel approaches to address identified challenges and advance the research agenda.

To do this, the Action will use networking and knowledge exchange to attract new partners and form cross-sector consortia, facilitating the formation of at least five multidisciplinary consortia by the end of Year 4, to drive innovation and advance the state of the art in healthcare simulation.

CBO 5. Establish a sustainable career trajectory for YRIs to develop the unique skills required to conduct innovative research in healthcare simulation (Addresses: C5)

This objective aims to cultivate the next generation of simulation researchers through targeted training, mentorship, and mobility opportunities across Europe. SEARCH will provide YRIs with structured pathways to develop their careers, fostering cross-sector and cross-geographical mobility and expanding the community of doctoral-trained researchers and innovators contributing to this field.

To do this, the Action will engage at least 50 YRIs in structured training and mentorship, including at least four fellowships, and ensure that around 30% of Action leadership roles are held by YRIs by Year 4.

C. State-of-the-art

Research advances over the past two decades have revolutionised our approach to safe and effective patient care, and to educating health professionals. With populations and environmental conditions changing and healthcare becoming increasingly complex, HCPs require more than individual knowledge and skills; they must learn to be able to work with colleagues across professions to ensure safe patient care (Brogaard et al., 2019). While safety has long been an implicit goal, it has now become a central focus of the healthcare quality improvement movement. Healthcare education has moved from lecture-dominant approaches toward experiential education through healthcare simulation, focused on how HCPs apply and develop their knowledge and skills (Frank et al., 2010). Healthcare simulation—techniques used to recreate specific events or settings for the purposes of training, assessment, quality improvement, or research—is an integrated approach that has shown great promise to enhance team and individual skill development, improve the quality and safety of care, and to research complex processes integral to safe patient care.

Healthcare simulation prepares HCPs to tackle complex healthcare challenges in modern team-based healthcare settings, and research is beginning to show benefits and highlight the areas of future focus. Simulation-based education (SBE) trains and develops a wide range of clinical, teamwork, and communication skills (e.g., Wayne et al., 2008; Shi et al., 2021). It enables learners to practice essential competencies such as patient interviewing, clinical examination, and performing interventions, as well as to rehearse the teamwork and communication skills required for managing complex patient care, including life-threatening conditions. Simulation also supports holistic care that addresses both patient and professional wellbeing (El Khamali et al. 2018), as emphasised in the WHO Global Patient Safety Action Plan 2021–2030 (WHO, 2021, p.49). Teams can rehearse complex procedures (such as multi-organ transplants) or refine time-critical processes like stroke management to minimise harm (Nickson et al., 2021). Simulation is also used to test new facilities, technologies, and systems, ensuring safe implementation and effective design.

Researchers can use simulation as a tool to better understand how clinicians care for patients.

Researchers can investigate either individual or team performance, and how health systems respond to complex care needs, such as emerging infectious diseases like COVID-19. By simulating healthcare settings, scenarios, and care pathways, researchers and quality improvement experts can identify and correct potential problems before they impact patient care. Adverse events can be forensically studied, and simulation exercises implemented to prevent their reoccurrence. Excellent patient care can be

explored in detail, to understand better how to train high-performing HCP teams. New treatments, processes, and multi-agency responses can be tested and refined before real-world application.

Many approaches and technologies can be used to enable or create simulated healthcare environments. Simulation modalities include computer-based simulations, procedural skills training with task trainers, mannequin-based simulation, simulated patient (using actors to portray patients), and Extended Reality (XR), which frequently are combined for improved realism (Lopreiato & Sawyer, 2015). XR, which includes virtual reality (VR), augmented reality (AR) and mixed reality (MR), holds significant potential to transform HCPs education and practice. Artificial intelligence (AI) is increasingly integrated with these technologies, creating adaptive, interactive learning systems that personalise training and assessment. AI-enhanced simulations generate highly realistic, interactive and culturally sensitive simulations that are scalable and adaptable across languages, cultures and resource settings (Rama & Caruso, 2025).

Simulation practice in healthcare rapidly across Europe and globally. In response to this expansion, simulation educators, researchers and scholars have formed professional societies to help elevate SBE at national (e.g., SPSIM in Portugal, SIMMED in Italy, DSSH in The Netherlands), transnational (ASPiH), European (SESAM), and international (SSH). While these societies share knowledge and practice, coordinated research endeavours are only just beginning to gain traction.

Networks of research, scholarship, and practice are emerging in other geographic areas, but are highly dependent on funding. North American-based research networks, for example, have built strong philanthropic support but largely focus on resuscitation science, reinforcing a narrow view of simulation as mainly for emergency care (Cheng et al., 2018). Broader applications, such as primary, palliative, mental, or social and cultural dimensions of care, remain underrepresented.

Approach to the challenges and advancements beyond the current state of the art

This Action seeks to address the thorny challenges presented in section 1.A by establishing a unique pan-European community of healthcare simulation research and innovation called SEARCH (Simulation Education And Research Collaborative in Healthcare). The main aim of this Action is to develop and establish a significant critical mass of academic, clinical, and industry stakeholder partners working toward a coordinated, sustainable, inclusive agenda of simulation research, education, and practice to improve healthcare in the European Research Area and beyond.

To address C1, rapid technological advancement, SEARCH will unify stakeholders from industry, academia, and clinical healthcare to continuously evaluate and respond to emerging technologies in healthcare simulation. Beyond established modalities, the Action will focus on cutting-edge technologies including digital twins, haptic feedback systems and AI-driven analytics. The integration of AI into simulation platforms enables adaptive and interactive learning experiences that personalise training and assessment. SEARCH will lead the development of frameworks and standards for responsible adoption, ensuring accessibility, cost-effectiveness, and impact on healthcare quality. This approach will help ensure that emerging technologies not only advance simulation but also transform healthcare practice. Maximising the utility, acceptance, and impact of AI and other emerging digital innovations to improve healthcare systems, increase quality, and enhance training will be a central focus of SEARCH.

SEARCH will address C2 and C3, focused on simulation for educating HCPs for dynamic healthcare environments and for improving healthcare quality and safety, by creating and nurturing sustainable research and knowledge exchange structures. These structures will make existing research evidence accessible through a public toolkit and support new multi-centre studies designed to generate high-quality data on simulation's impact on education, quality and safety. SEARCH will identify the most pressing research questions and build a collaborative network with the relevant expertise to address them. This coordinated approach will enable rigorous evaluation of emerging simulation technologies and methodologies, demonstrating their fitness for purpose in transforming and improving healthcare, thus substantially advancing the state of the art.

Shifting from traditional ways of working in healthcare, **SEARCH will address C4, breaking down siloes and encouraging collaboration, by creating a platform for stakeholders to connect, work, learn, and co-create new knowledge.** The field of healthcare simulation has evolved from diverse disciplines including medicine, nursing, safety science, behavioural science, engineering, computer science, and aviation—each with its own established community. As a result, building a shared, cross-disciplinary Community of Practice (CoP) has been inherently challenging. The international and interdisciplinary nature of SEARCH will break down geographical, linguistic, professional, and disciplinary silos,

connecting both novices and experts from complementary domains to jointly address current and future healthcare challenges. By creating sustainable and self-governing coordination mechanisms, SEARCH will map the evolving research and technology landscape across Europe and beyond, ensuring that new initiatives complement rather than duplicate existing efforts. In doing so, SEARCH will strengthen collaboration, accelerate innovation, and contribute to improving healthcare quality and safety.

To address C5, nurturing a new generation of simulation researchers, **SEARCH will establish clear and sustainable career trajectories for colleagues across industry, academia, and clinical practice to develop themselves as healthcare simulation researchers and practitioners.** SEARCH will build a coherent and structured approach to research training and capacity building, creating opportunities for emerging leaders to take on leadership and coordination roles, especially for YRIs and colleagues from ITCs. In parallel, SEARCH will identify and engage experienced professionals to act as mentors, supporting career development. By developing a viable and sustainable simulation research community, SEARCH will also facilitate mobility between industry, healthcare, and academia, enhancing both professional growth and cross-sector collaboration across Europe and beyond.

D. Rationale for choosing networking to address the main challenge

The international simulation research community remains highly fragmented, preventing the sector from engaging with and contributing to real-world healthcare challenges. Currently, simulation research and innovation are driven by isolated hubs of field-specific expertise that often pursue disconnected priorities. Within Europe, these hubs include aforementioned national, transnational, and pan-European societies. Internationally, networks and societies such as INSPIRE, SSH, AsiaSim, and GNSH play similar roles. While these organisations have successfully fostered expertise in simulation education, they lack a unified research agenda and a coordinated mission to advance shared priorities. As a result, leadership within the field remains fragmented, causing research, technology development, and practice to advance in an uncoordinated manner. This fragmentation impedes progress across Europe, as duplication of effort, overlapping priorities, and limited career pathways continue to hinder collaboration and constrain innovation. Consequently, research findings often remain confined to local contexts, reducing their potential to address broader healthcare problems and to drive systemic improvements in healthcare quality and safety. SEARCH will therefore act as a “network of networks”. By building on and connecting existing structures, SEARCH will identify and avoid duplication of work, set a shared research agenda, and harness Europe’s collective expertise towards shared goals. Bringing together industry leaders, researchers, policymakers, and practitioners, SEARCH will ensure that innovation, research, and policy are not only aligned but embedded within decision-making processes. This integrated approach will accelerate the translation of research into actionable, evidence-based solutions, driving systemic change in healthcare simulation practice across diverse contexts. In turn, this will promote innovation, facilitate evidence-based practices, and improve equity of access to knowledge and skills, benefiting patient outcomes across Europe and laying the groundwork for a sustainable, responsive, dynamic community that will continue well beyond the life of the Action.

This Action will add value to the European healthcare simulation landscape by building a cohesive European simulation research community of practice and creating networking opportunities that promote the exchange of excellence in research and practice. This Action will facilitate the translation of simulation research to evidence-based practice, empowering partners to recognise and adopt innovative technologies and methodologies that benefit their local healthcare systems. SEARCH will actively engage with existing European networks, simulation industry leaders, and policymakers in Europe. Emerging networks in ITCs will be integral members, ensuring that expertise and capabilities are spread across the continent. The Action will also promote the representation and participation from COST Partner members and International Partners, fostering global knowledge exchange and amplifying the impact of European simulation research worldwide. By connecting partners across geographical, linguistic, and disciplinary boundaries, SEARCH will bridge existing divides and promote standardised yet adaptable approaches suited to diverse healthcare contexts. In doing so, it will establish a coherent and sustainable European simulation research ecosystem that continues to advance and generate impact well beyond the life of the Action.

This Action will recruit members with a broad spectrum of methodological expertise, disciplinary backgrounds, and cultural perspectives, capturing diverse geographic, linguistic, and other nuances. SEARCH will draw on the knowledge of multidisciplinary and multi-agency HCPs experienced in facilitating simulation practice and faculty development, representing both contributors to and

beneficiaries of the network. This will enable coordinated research and capacity-building activities across Europe, supporting sustainable career trajectories for YRIs as well as experienced professionals transitioning into healthcare simulation. Through this coordinated effort, SEARCH will create and nurture the structure and strategies required to enhance European simulation expertise in research skills; to improve translation of existing evidence from research into simulation practices; to augment professional, cultural and organisational collaboration; and – most importantly – to improve patient care.

E. Critical mass of the network

This Action will bring together a diverse group of experts with complementary experience, expertise, and perspectives in empirical and applied simulation research, with a strong emphasis on inclusivity and sustainable development of the field. To address the critical challenge of unifying and advancing simulation research and innovation across Europe, the Action is designed to secure the necessary critical mass of expertise and achieve a balanced representation. The initial consortium already embodies significant intersectoral collaboration, involving leading academic institutions, research centres, hospitals, and industry partners. It includes scholars and researchers with formal training and expertise across key domains such as simulation technologies, computer science, organisational behaviour, team science, health professions education, and clinical health sciences. This multidisciplinary foundation ensures that the network is well equipped to address the full range of challenges in advancing healthcare simulation research and practice. Proposers of this Action view this constellation as an initial critical mass in terms of expertise, experience, and geographic diversity, which will be expanded through the social practice logic of innovation spread and scale-up (Greenhalgh & Papoutsis, 2019) to broaden participation, extend impact, and ensure long-term sustainability of the field.

The proposers will work diligently to expand and extend this initial critical mass, by leveraging existing partnerships and developing new collaborations to forge and extend SEARCH. This Action will foster connections with simulation leaders active in national and international simulation societies, which, although primarily focused on simulation practice and professional education, are key stakeholders in advancing simulation research. This Action will also invite and collaborate widely with industry partners, technology developers and patient representatives to ensure the network reflects the full value chain, from innovation through implementation to impact on care. Involving experts in standardisation, regulatory frameworks, and policy will further strengthen the network's ability to translate research into real-world practice.

By strengthening excellence through the creation of cross-border researcher networking, the Action will increase their efficiency, effectiveness, and impact at a European and global level. Mentorship opportunities for YRIs, supported by experienced and active simulation researchers, will be extended and embedded through the Action and will facilitate creation of a mutual learning infrastructure and community of practice. The principles of Participatory Research will underpin the network, specifically “the co-construction of research through partnerships between researchers and people affected by and/or responsible for action on the issues under study” (Jagosh et al., 2012, p. 312). SEARCH will connect stakeholders from diverse professional, geographic, and disciplinary backgrounds to foster dialogue on key challenges affecting healthcare workforces across Europe and to co-develop innovative, evidence-based solutions.

2. IMPACT

A. Impact related to objectives

This Action will establish scalable and sustainable initiatives to foster a European-wide cross-disciplinary culture of exchange and cooperation in healthcare simulation science, advancing the state of the art. SEARCH will address the fragmentation of Europe's simulation research ecosystem through network building, coordinated knowledge exchange, a long-term research agenda and capacity-building. This will support effective healthcare delivery and patient safety, strengthen the European Research Area, and enhance educational, organisational, and cultural developments. Together, these objectives will generate the following impacts, addressing specific challenges (C).

Short-term impact	Long-term impact
Educational Impacts	
• Development of local and pan-European knowledge of 'What Works' repository in	• Reduced barriers to access for high quality and locally relevant healthcare education across

simulation education through research and dissemination activities. (C2) • Sustained networking for YRIs (or researchers new in this field) and ITC members, with international research collaborations and proposals emerging from the Action. (C5) • Creation of infrastructure to support future generations of healthcare simulation researchers and practitioners in academia, healthcare, and industry, including a sustainable career trajectory for YRIs. (C5) • Co-creation of educational standards for dynamic, mobile, multilingual, and culturally diverse healthcare force (C2).	geographies, professions, and technology access. (C2) • Extended scope of healthcare simulation research and education to encompass the evolving community of researchers and disciplines within the sector, contributing to innovation in research in the field. (C5) • Increased participation of YRIs in simulation research, with a focus on increased capacity in underrepresented countries. (C5) • Standardisation and identification of best practice (e.g., terminology, competencies) to ensure consistency and quality. (C1, C3, C4)
Scientific Impacts	
• Co-creation of a clear agenda for the next decade of knowledge generation, practice, and research in the field of healthcare simulation, guided by co-produced and inclusive priorities across the ERA. (C4) • Establishment of a robust network of researchers who plan future grant applications for collaborative, large-scale, multi-site research in Horizon Europe. (C4) • Establishment of a Healthcare Simulation Research Thematic Network in the EU Health Policy Platform, bringing together different stakeholders to address geographical and disciplinary silos, amplify and share expertise, and coordinate continued capacity building efforts for research excellence. (C4)	• Increased multidisciplinary research output collaboration across technology, healthcare, and education experts to respond to emerging healthcare challenges. (C2, C3, C4) • Accelerated progress in the field's state of the art through continued alignment of research priorities to meet evolving healthcare needs. (C1, C2, C3, C4) • Increased cross-country collaboration to solve pan-European healthcare challenges through simulation research. (C4) • Established pathways to facilitate mobility of researchers between industry, healthcare, academia, and geographical locations for seamless knowledge translation and implementation. (C2, C3, C5)
Technological Impacts	
• Development of standards for effective, sustainable technological innovations, including emerging technologies. (C1) • Understanding of pathways and evidence required to facilitate the implementation of new simulation technologies in education, quality improvement, and patient safety. (C1, C2, C3) • Understanding of the application of simulation technologies to address healthcare education, research, and practice challenges. (C1, C2, C3) • Identification of technological gaps and barriers for implementation. (C1)	• Consolidated partnerships between industry, research, and healthcare to improve the responsiveness to changing healthcare needs through technological innovation. (C1) • Acceleration of adoption of new immersive technologies through research and industry partnerships to address inequities in simulation access across Europe. (C1) • Coordination of pan-European multi-centre studies to develop high quality evidence for the implementation and innovation in new effective technologies. (C1, C2, C3, C4)
Socio-economic Impacts	
• Creation of sustainable pathways for integrating patient voices into healthcare simulation research and practice. (C2, C4) • Identification and mitigation of barriers that can impede effective teamwork and safe patient care — including workforce mobility, linguistic challenges, and professional silos — through healthcare simulation. (C2, C3) • Application of implementation frameworks to translate findings from simulation research into healthcare practice. (C2, C3) • Continuous review of most effective approaches in simulation education for improved resource	• Embedded culture of co-production with patients and the public in simulation research and healthcare practice. (C2, C3, C4) • Improved teamworking in healthcare with impacts on health quality and outcomes for patients and HCPs. (C3) • Reduction in healthcare inequalities through the equitable provision of high quality and evidence-based simulation training across Europe. (C1, C2, C4) • Embedded and standardised utilisation of simulation research within healthcare quality improvement processes in Europe to align sector

allocation and cost- efficiencies in healthcare education. (C3)	wide practice and decision-making with scientific evidence-base. (C1, C3)
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Impact on research coordination and capacity building

To maximise its relevance and effectiveness, SEARCH will prioritise three overarching outcomes, each operationalised through a series of interrelated and synergistic activities directly related to the objectives of this Action (RCO and CBO).

Knowledge creation

- The Action will create knowledge by identifying barriers and facilitators for healthcare simulation research and practice through a structured needs analysis (CBO 2). Working Groups (WGs) will establish coordinated processes to gather data on healthcare SBE activities, considering the specific organisational, sociocultural, equity, and health contexts in the wide variation of communities the Action serves, ensuring voices from marginalised and underserved communities are represented. This will include surveying at least 50 simulation healthcare facilities, local health policy makers, and patient representatives across Europe and beyond to determine and document current practices and challenges in simulation training.
- Guided by the data gathered, WGs will collaboratively design innovative healthcare simulation approaches aiming at enhancing patient safety, improving the quality of healthcare delivery, and strengthening patient engagement through better education and training (RCO 2). At least 10 simulation-based activities will be designed as a showcase of how simulation can be used for both educational and translational purposes. These activities will include adaptations for low-resource settings and culturally diverse contexts, and local pilot sites will be encouraged to share early implementation experiences.
- SEARCH will generate a comprehensive research agenda (RCO3) and a series of white papers, which will be produced and published during the lifespan of this Action, with recommendations for how healthcare simulation can contribute to effective team working, patient safety, enhanced overall quality of care, individual HCP resiliency, resilience of health systems, and improved patient and health policy makers engagement and partnership. These initiatives will advance the field by proposing strategies for equitable access to simulation resources, training and infrastructure.

Knowledge transfer, mobilisation and application

- A SEARCH launch event will mark the beginning of the Action and promote its activities and programme. This event will serve as a critical platform for engaging key stakeholders identified in section 2.B, aiming to solidify and expand the network. Leveraging the extensive experience of the Action partners in fostering the transfer of knowledge between industry, the academic/clinical sectors, health policy makers and patient representatives (RCO2), this event will set the stage for collaborative engagements throughout the Action.
- A community-engaged SEARCH Toolkit will include an interactive, modular, and curated online platform. This platform will host all produced materials by the Action, ensuring accessibility for Action partners, external experts/audits, policy makers, patient representatives, and general public (RCO2 and CBO1), with regular updates during this Action. The online platform will also promote the Actions and WGs activities and progress. The Toolkit will remain available beyond the lifespan of the Action, promoting continuous knowledge transfer and serve as a long-term resource for the community.
- The Short-Term Scientific Missions (STSMs) and Training Schools held by international experts will allow focused knowledge transfer to YRIs, researchers, scholars, and healthcare practitioners (RCO1, RCO2, CBO3, CBO5). A minimum of 4 training schools and 6 STSMs opportunities (half of those in partnership with industry) will be promoted through the Action lifespan. Additionally, 12 online Masterclasses will be held. At least, 4 Fellowships will be supported during the Action.
- SEARCH will facilitate the establishment of local communities of practice where practitioners can exchange knowledge and receive mentorship during the early stages of simulation initiative implementation (RCO1, CBO3). Research outcomes will be disseminated through conference presentations, network events, workshops, and open-access publications. These activities will support the growth of local communities of practice and attract new members to the Action. Workshops will be hosted across different regions, with priority given to ITC partners. Each year, at least two workshops and one network event will be integrated into major international conferences, serving as platforms for knowledge exchange and as opportunities to recruit new Action members (CBO3, CBO4).
- SEARCH showcase event will conclude the Action activities. This event will focus on celebrating the

achievements and ensuring that the network and collaborations continue beyond the Action (RCO1).

Career development

- This Action targets the training and development of a multi-professional audience, ranging from YRIs, researchers new to this field, HCPs, and others (RCO1, CBO5). The WG that will focus on capacity-building and skills development specifically aims to promote targeted development of these audiences. Educational practical courses and dedicated Training Schools, with programs focused on healthcare simulation research and practice will be organised at least twice a year (face-to-face or online to provide equal access) (RCO1, CBO3). These initiatives will be designed for sustainability, so they can continue after the structures of the network are established (RCO1, CBO1).
- YRI grants and fellowships will allow novice researchers to gain visibility at international conferences (CBO4, CBO5). STSMs will be organised, stimulating collaboration and mentoring for YRIs or researchers new to this field. YRIs and ITCs will be encouraged to hold Action leadership positions to strengthen their careers, develop capacity in ITCs and links between ITCs and other COST members, and develop diverse perspectives to shape the field in the future (RCO1, RCO2, CBO5).

B. Involvement of stakeholders

SEARCH will implement a Quadruple Helix approach (involving academia, industry, government/policy, and civil society) to engage a broad and diverse range of stakeholders. At the network's launch, an initial stakeholder mapping exercise will identify existing partners and potential collaborators, with particular attention to underrepresented groups. Stakeholder engagement will be coordinated by the Management Committee, supported by a Stakeholder Liaison Coordinator who will work closely with a group of key stakeholders experienced in simulation to recruit a diverse cohort of emerging stakeholders (see Table below). All participants will be committed to the principles of scientific integrity, collaboration, equality, and to shaping the strategic direction of healthcare simulation research.

Academia (universities; research institutes)

Roles and contributions: Provide disciplinary expertise across simulation technologies, pedagogy, cognitive psychology, organisational behaviour, engineering, computer science, health sciences and change management. Develop research methodologies, contribute to WGs, co-author reports/white papers and train the next generation.

Engagement plan and challenges: Involved from the start as proposers. Annual academic roundtables and thematic workshops will ensure cross-disciplinary collaboration. Participation in all WGs, leading methodological development and research advancement. Challenge: coordinating diverse research cultures. Mitigation: terms of reference and facilitation support.

Industry

Roles and contributions: Manufacturers and developers of simulation hardware/software (manikins, VR/AR, AI platforms, digital twins), SMEs and start-ups. Participation in all WGs: co-develop standards, identify technological opportunities, provide technical experts for WGs, offer placements for STSMs and fellowships, and disseminate innovations.

Engagement plan and challenges: Industry representatives involved from the start as proposers. Additional representatives invited during Year 1. Challenge: balancing commercial interests with open research. Mitigation: transparent IP agreements.

Healthcare (hospitals, primary care, social care organisations)

Roles and contributions: Supply real-world clinical challenges and data, integrate simulation into education and quality improvement programmes. Ongoing involvement by participating in all WGs.

Engagement plan and challenges: Involved from the start as proposers. Additional engagement during Year 1 with the establishment of local Communities of Practice. Challenge: time and resource pressures. Mitigation: flexible virtual participation and targeted support for smaller/remote centres.

Simulation Centres

Roles and contributions: Provide dedicated facilities and technical expertise for simulation training and research, including access to equipment, experienced facilitators and simulation technicians. Pilot and evaluate new methodologies, share operational best practices and host training schools and STSMs. Act as hubs for local CoPs, supporting early implementation and feedback loops.

Engagement plan and challenges: Included as core partners from the outset; surveyed during Year 1 needs analysis. Participate in all WGs to inform research design and implementation. Host regular SEARCH events and offer placements for STSMs and fellowships. Challenge: variability in

infrastructure, staff capacity and funding across centres. Mitigation: tailor support (e.g., virtual training, shared resources); encourage peer-to-peer mentorship between resourced and emerging centres.

Policy analysts and government/regulatory bodies

Roles and contributions: Policy analysts, government, accreditation agencies and professional bodies. Translate research findings into policy, support regulatory harmonisation and accreditation, and identify funding mechanisms. Provide strategic oversight through an advisory panel.

Engagement plan and challenges: Invited during Year 1 to define research–policy interface. Annual policy briefs and targeted workshops will support engagement. Challenge: differing national priorities. Mitigation: align recommendations with European agendas (e.g., patient safety directives) and highlight benefits for workforce planning.

Scientific and professional societies

Roles and contributions: European, trans-national, and national simulation societies, quality and safety agencies, patient safety organisations and non-governmental organisations with an interest in simulation. Help disseminate results, align standards and expand the network.

Engagement plan and challenges: Included from the outset through Memoranda of Understanding. Participate in annual conferences and co-host dissemination events. Challenge: varying missions and membership structures. Mitigation: establish liaison points and cross-promotion.

Patients and public representatives

Roles and contributions: Provide lived experience, define patient-centred priorities, co-produce research design and evaluation, and ensure inclusivity. Assist with scenario development, technology evaluation and dissemination to civil society.

Engagement plan and challenges: A Patient & Public Involvement (PPI) group will be established in Year 1. Members will participate in WGs and be invited to co-create workshops. Challenge: ensuring meaningful engagement without tokenism. Mitigation: use facilitators experienced in PPI, reimburse expenses, and provide training on research processes.

C. Communication, dissemination and valorisation

The Action will implement a structured communication, dissemination and valorisation strategy to reach diverse stakeholders. Action dissemination activities will be coordinated by the Science Communication Coordinator, who will join WG1 and work in close collaboration with other WG leaders and the Management Committee (MC) and in alignment with the detailed dissemination plan to be agreed in WG1. It will detail the objectives, audiences, channels, responsibilities and metrics, and will be reviewed annually to ensure it remains responsive and effective. Dissemination activities will also aim to extend the network, generate interest in the project amongst YRIs, share progress with stakeholders and share outputs with the scientific community and the public.

Audiences and targeted activities

- **Research community and YRIs:** progress reports, research findings and methodological advances will be disseminated through peer-reviewed publications, open-access repositories, webinars, training schools and conference sessions.
- **Healthcare practitioners and educators:** practical guidance, case studies and training materials will be shared via interactive workshops, simulation forums and regional seminars; content will be multilingual and accessible through the online platform.
- **Industry and technology developers:** targeted briefings, roundtables and bilateral-meetings will facilitate collaboration on new simulation technologies, ensuring industry needs and innovations inform research priorities.
- **Policymakers and regulators:** policy briefs, white papers and dedicated workshops will present evidence from the Action and highlight implications for workforce training, patient safety and health system resilience. The Action will establish a Healthcare Simulation Research Thematic Network within the EU Health Policy Platform to provide a standing forum for policy engagement.
- **Civil society and patient organisations:** plain-language summaries and infographic, public webinars and engagement at outreach events will explain how simulation research benefits patients and address local needs.

Communication channels and activities

- **Web-based platform and SEARCH Toolkit:** a public, multilingual platform will host all deliverables, including reports, protocols, best-practice guidelines, educational materials and videos. It will include an interactive forum to support networking and peer-to-peer learning. The platform will remain active

beyond the Action's lifespan, with stewardship transferred to partner organisations and the Representative Council.

- **Social media:** Coordinated campaigns will be used to maximise impact, targeting healthcare and simulation communities at local, national and international levels. A diverse approach will include professional networks (LinkedIn), community platforms (Bluesky) and video channels (YouTube).
- **Conferences and Roundtables:** Conferences and roundtable discussions will serve as key platforms for disseminating research findings and fostering industry and academic collaborations. These events will be open to researchers and innovators (both YRIs and experienced), HCPs, and the public, and local health policy decision-makers and relevant stakeholders, such as industry and patient groups, will be invited. Annually, one network event will be held at a major European simulation conference (e.g., SESAM).
- **Publications and open science:** In the spirit of equity, publications from Action activities will be freely accessible to all by targeting journals with an Open Access license. All outputs will be widely disseminated through the network and published openly on the web platform. Participants will be encouraged to submit to the Commission's Open Research Europe (ORE), the Open Access publishing platform. At least 5 papers in high-impact open-access journals, such as *Advances in Simulation*, will be published by the end of the Action.
- **Dissemination workshops/seminars:** Local communities will be invited to join virtual and face-to-face events such as interactive workshops, seminars, and lectures to disseminate knowledge and to maintain an open dialogue between the Action partners and the broader public. Each workshop/seminar will target a minimum of 25 participants.
- **Policy-maker workshops/roundtables:** Regional/national policy makers will be invited to attend dedicated workshops and roundtables, promoting awareness and policy changes sustained by the research outputs from the Action WGs. Open dialogue with local policy makers will be fostered, and the policy makers will be encouraged to consider research in the future policies and curricula in various partner countries. Further, this Action will propose Healthcare Simulation Research Thematic Network in the EU Health Policy Platform (HPP) during the lifetime of the Action.
- **Active on-campus dissemination:** To bring greater awareness of the Action amongst students and young researchers, the Action partners will take an active role in the dissemination of the Action outcomes and activities in their organisations. Throughout the Action lifespan, at least 10 partner institutions will host a local event with a target audience of at least 100 participants.

Intellectual Property and Exploitation: The Management Committee (MC) will develop clear guidelines for managing intellectual property rights arising from the Action, ensuring that all partners' interests are protected, including acknowledgement of COST. Results that have commercial or practical application will be identified and, where appropriate, subject to exploitation plans in partnership with institutions' research and innovation offices. The Action will pursue a COST Innovators' Grant and explore collaborations with industry to develop and scale promising outputs (e.g., software tools, training modules or technology standards) while maintaining the core ethos of open access and public benefit.

3. IMPLEMENTATION

A. Action Structure

Action coordination and management: The Action will be coordinated by the Management Committee (MC), operating through a flat and inclusive governance model that promotes shared decision-making across partners, Figure 1. Leadership roles will be elected in line with COST Rules and Principles. Principles of equity, diversity, and inclusion (EDI) will guide the MC, with explicit goals of gender balance, at least 30% representation of YRIs, and at least 50% representation from ITCs.

The MC is responsible for overall coordination, implementation, monitoring, and impact assessment of the Action, as well as transparent allocation of COST funding to meet strategic objectives. It will

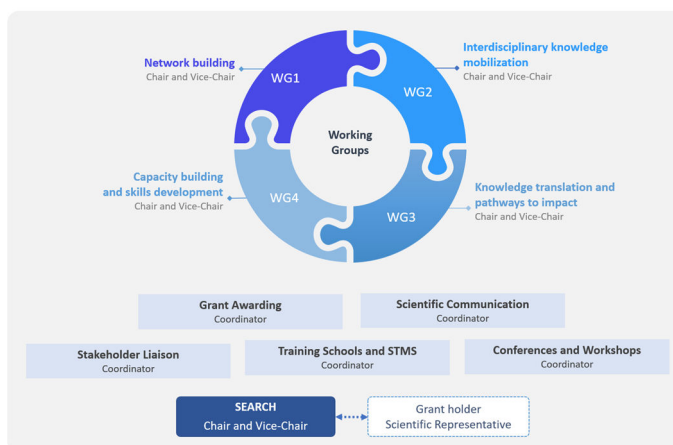


Figure 1. SEARCH MC Composition

meet twice annually in hybrid format, with additional virtual meetings as required. To support efficiency and continuity, a Core Group, comprising the Chair, Vice-Chair, Working Group Leaders, the Grant Holder Scientific Representative, the Science Communication Coordinator, and the Grant Awarding Coordinator—will manage day-to-day matters between MC meetings. Additional supporting roles (Stakeholder Liaison Coordinator, Training Schools and STSM Coordinator, Conferences and Workshops Coordinator) will ensure effective stakeholder engagement and smooth delivery of training and dissemination activities. The MC will have dedicated project support, including a project manager, finance team, and research support team with experience of managing large European Union-funded research and innovation grants. The MC will closely support WGs, which will meet quarterly to develop and implement their workplans. The WG Chair and Vice-Chair will oversee workplan execution, monitor activities, and report to the MC. Based on the recommendations of all Coordinators, the MC will plan the annual budget and define the activities.

Working groups: The Action will be implemented through **four interdependent WGs**. Each WG has clear objectives, tasks, and deliverables, and will ensure representation across disciplines, sectors, and generations. Cross-WG collaboration will be embedded through joint activities and shared evaluation processes, ensuring coherence and synergy across the Action.

- **WG1. Network Building** – Establishes the Action’s foundation by connecting people, infrastructures, and knowledge across Europe. Provides the networking infrastructure (platform, Special Interest Groups, events) that supports and enables the activities of all other WGs.
- **WG2. Interdisciplinary Knowledge Mobilisation** – Promotes multi-directional knowledge exchange across research, practice, industry, and education. Ensures evidence and best practices are widely accessible, while feeding into WG3’s agenda-setting and translation activities.
- **WG3. Knowledge Translation and Pathways to Impact** – Translates simulation research into policy, practice, and system-level improvements. Develops long-term research agendas and impact pathways that inform WG4’s training and skills priorities.
- **WG4. Capacity Building and Skills Development** – Builds research capacity and diversifies the field by training and empowering YRIs, strengthening interdisciplinary collaboration, and fostering sustainable career trajectories. WG4 builds skills and develops the next generation of researchers who will contribute back to WGs 1–3, ensuring sustainability.

B. Work plan (tasks, activities and timeframe)

WG1. Network building (M1-48) Addresses: RCO1, CBO1, CBO4	
Tasks	T1.1 Develop sustainable and lasting communication and governance structures T1.2 Build and maintain the SEARCH platform T1.3 Network growth and member engagement T1.4 Continuously map expertise and infrastructure T1.5 Dissemination and outreach
Activities	• Draft the communication and dissemination strategy and establish a Representative Council • Develop a web-based, open-access platform/repository of resources • Form special interest groups (SIGs) with topical priorities for the wider network • Organise networking events at existing conferences • Conduct a survey to document resources, expertise and gaps. • Create appropriate dissemination channels
Timeline Milestones	M1–M6: Strategy drafted and approved; M12: Representative Council formed. M1–M48: Platform development and annually updates; M12: platform launched; M4–M48: Targeted membership campaigns; M12: network reaches 80 members; M24: network meets 150 members; M48: ≥300 members. M7–M48: Resource mapping and annually updates; M18: Resource map published; M7–M48: Dissemination activities; M48: Final dissemination report.
WG2. Interdisciplinary knowledge mobilisation (M7-48) Addresses: RCO2, CBO1, CBO2	
Tasks	T2.1 Conduct needs analysis and identify best practices T2.2 Identify barriers and enablers for simulation research T2.3 Disseminate knowledge and share best practices

Activities	• Conduct a network needs analysis to identify knowledge and skill gaps • Synthesise identified gaps, barriers and opportunities to inform the WG3 research agenda; • Develop a report with barriers, opportunities, and mitigation strategies; • Compile 10 case studies on successful knowledge mobilisation/translation; • Deliver 8 online/hybrid seminars covering priority topics identified in the needs analysis; • Update the Toolkit with protocols, reports and tools
Timeline Milestones	M7–M24: Survey design, data collection, data analysis; <i>M18:</i> needs analysis report M25–M48: Case study collection; <i>M30:</i> barriers and opportunities report released. <i>M48:</i> Report on best practices. M13–M36: Seminars at M15, M18, M21, M24, M27, M30, M33, M36; M13–M48: Regular contributions to agenda drafting (WG3); <i>M36:</i> final input incorporated.
WG3. Knowledge translation and pathways to impact (M13–48) Addresses: RCO3, CBO2, CBO3	
Tasks	T3.1 Build capacity across the SBR community in implementation science T3.2 Develop coordination structures for knowledge translation T3.3 Share guidance and best practices in knowledge translation for SBR T3.4 Co-create research agenda and white papers
Activities	• Design and deliver 4 workshops on implementation science. • Establish local CoPs in diverse regions to facilitate knowledge translation. • In partnership with WG2 and stakeholders, draft a ten-year European research agenda addressing technological advances, education, quality and safety. • Develop guidelines for translation processes • Propose a Healthcare Simulation Research Thematic Network in the EU Health Policy Platform.
Timeline Milestones	M13–M48: Workshops at M15, M21, M27, M33; <i>M48:</i> workshop materials archived in Toolkit. M13–M48: Initial COPs established by M18; M25–M48: Advisory group formed; <i>M36:</i> guidelines published; M25–M48: Thematic Network proposal submitted and operationalised. M19–M48: Stakeholder consultations; <i>M42:</i> agenda published; <i>M48:</i> white papers released. M13–M48: Policy roundtables at M18, M30, M42; <i>M48:</i> policy recommendations submitted.
WG4. Capacity building and skills development (M1–48) Addresses: CBO3, CBO4, CBO5	
Tasks	T4.1 Develop curricula and training materials to address identified skill and knowledge gaps T4.2 Deliver training programmes T4.3 Coordinate STSMs, fellowships and YRI leadership T4.4 Evaluate and ensure sustainability
Activities	• Design a modular curriculum addressing identified skills and knowledge gaps • Prepare a syllabus, presentation material, and lecture notes • Provide at least four Training Schools (3–15 days) on priority topics; • Offer 12 online masterclasses; • Allocate 6 STSMs and 4 fellowships over the Action, prioritising YRIs and ITC participants. • Collect feedback from participants and stakeholders; conduct annual reviews to adapt training and COPs; develop guidelines for sustaining programmes beyond the Action.
Timeline Milestones	M19–M24: Curriculum drafted and reviewed; <i>M24:</i> curriculum published M25–M48: Masterclasses and training schools launched at M24; <i>M48:</i> training programme completed M1–M48: STSM and fellowship; <i>M48:</i> STSM and fellowships completed. M12, M24, M36, M48: Evaluation checkpoints with improvement actions; <i>M48:</i> sustainability plan delivered.

C. Deliverables

D1. Communication and Dissemination Strategy (WG1 T1.1)	M6
A written strategy describing target audiences, messaging, channels, metrics and responsibilities.	
D2. Governance Charter and Representative Council Terms of Reference (WG1 T1.1)	M12
Document defining governance, membership rules and procedures for the Representative Council.	
D3. SEARCH Platform and Toolkit – initial release (WG1 T1.2)	M12
Launch of the open-access platform with resource map integration and initial documentation.	

D4. Resource Map Report – initial release (WG1 T1.4)	M18
Report describing the mapping of simulation centres, expertise and resources across Europe.	
D5. Needs Analysis and Gap Identification Report (WG2 T2.1)	M18
Comprehensive report on knowledge/skills gaps, based on surveys of ≥50 centres and stakeholders.	
D6. Barriers and Enablers Report with Case Studies (WG2 T2.2)	M30
Report of barriers to knowledge mobilisation, including case studies and mitigation strategies.	
D7. Best Practices and Mitigation Strategies Compendium (WG2 T2.3)	M48
Final report compiling evidence-based best practices for simulation research and translation.	
D8. CoPs Establishment and Guidelines Document (WG3 T3.2)	M18
Guidance document for establishing local CoPs, including initial CoP reports and lessons learned.	
D9. Thematic Network Proposal and Implementation Plan (WG3 T3.2)	M30
Proposal of a Healthcare Simulation Research Thematic Network and a roadmap for implementation.	
D10. Guidelines for Knowledge Translation (WG3 T3.3)	M36
Practical guidelines for translation processes and coordination structures for simulation research.	
D11. European Research Agenda for Healthcare Simulation (WG3 T3.4)	M42
A co-created ten-year research agenda with governance and sustainability plan.	
D12. White Papers (WG3 T3.4)	M48
At least two white papers derived from the research agenda.	
D13. Policy Recommendations Report (WG3 T3.3)	M48
Report of the outcomes of policy roundtables, including actionable recommendations for governing and accreditation bodies.	
D14. Modular Curriculum and Training Material Package (WG4 T4.1)	M24
Modular curriculum and supporting materials addressing identified skill gaps.	
D15. STSM and Fellowship Outcome Report (WG4 T4.4)	M48
Report summarising the objectives, outcomes and career impact of six STSMs and four fellowships.	
D16. Sustainability Plan (WG4 T4.5)	M42
Sustainability plan for maintaining programmes post-Action.	
D17. Final Dissemination and Impact Report (cross-WG)	M48
Comprehensive report compiling annual evaluation reports, summarising dissemination activities, engagement metrics, research coordination outcomes, capacity-building impacts and policy uptake.	

D. Gantt chart

Tasks	Year 1				Year 2				Year 3				Year 4			
	1-3	4-6	7-9	10-12	1-3	4-6	7-9	10-12	1-3	4-6	7-9	10-12	1-3	4-6	7-9	10-12
Network Management																
Overall coordination and financial management of the Action																
WG1. Network building																
WG overall management (e.g., meetings)																
T1.1 Develop communication and governance structures			D1	D2												
T1.2 Build and maintain the SEARCH platform				D3				D3				D3				D3
T1.3 Network growth and member engagement		A							A							A
T1.4 Continuously map expertise and infrastructure						D4		D4				D4				D4
T1.5 Dissemination and outreach																
WG2. Interdisciplinary knowledge mobilisation																
WG overall management (e.g., meetings)																
T2.1 Conduct needs analysis and identify best practices						D5										
T2.2 Identify barriers and enablers for simulation research										D6						
T2.3 Disseminate knowledge and share best practices					A	A	A	A	A	A	A	A				D7
WG3. Knowledge translation and pathways to impact																
WG overall management (e.g., meetings)					A	A	A	A								
T3.1 Build capacity across the SBR community in implementation science																
T3.2 Develop coordination structures for knowledge translation						D8				D9						
T3.3 Share guidance and best practices in knowledge translation for SBR												D10				D13
T3.4 Co-create research agenda and white papers														D11		D12
WG4. Capacity building and skills development																
WG overall management (e.g., meetings)																
T4.1 Develop curricula and training materials								D14								
T4.2 Deliver training programmes									A		A		A		A	D15
T4.3 Coordinate STSMs, fellowships and YRI leadership		A	A	A	A	A			A	A		A		A		
T4.4 Evaluate and ensure sustainability				D17				D17				D17		D16		D17

Note: "D"s refers to Deliverables. For a description of each deliverable, please refer to Section 3C.

"A" refers to Activities (such as Network events, Seminars, Workshops, Masterclasses, Training schools, STSM, or Fellowships). For a complete description of the Action activities, please refer to Section 3B.

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